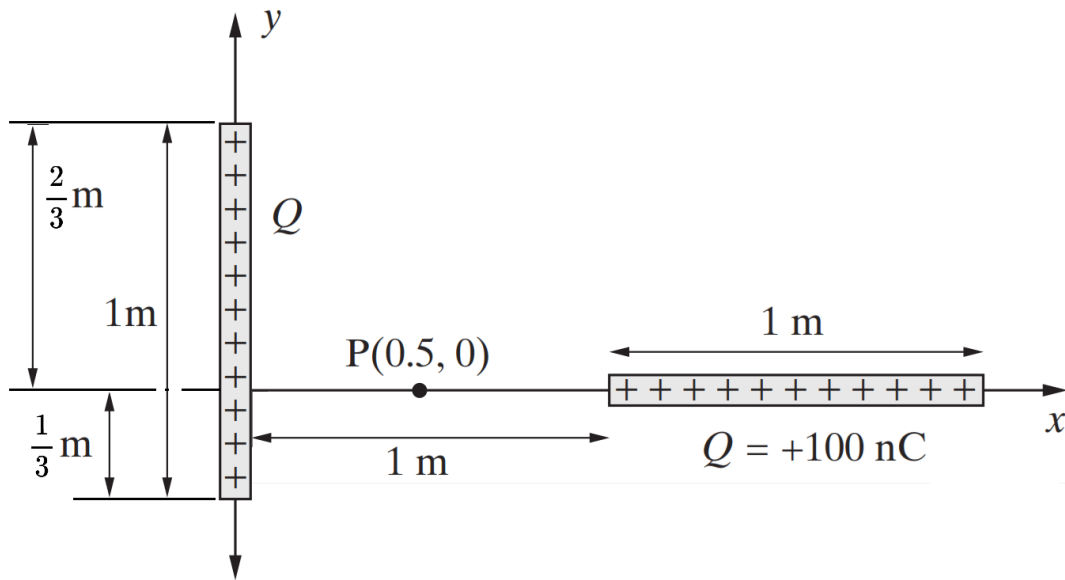




Q1: [20] Three point charges. Three point charges of values $+300\text{nC}$, $+200\text{nC}$, and $+400\text{nC}$ are located at points $(0, 0, 0)$, $(1, 0, 0)$, and $(0, 1, 0)$, respectively. Find the force on each charge due to the other charges. Which charge experiences the largest force?

Q2: [40] Consider two uniformly charged wires, each of length 1 m and a total charge $+100\text{ nC}$, with their ends separated by 1 m , as shown in Figure below.

- (a) Find the electric potential V at the point P , midway between the two wires.
(b) Find the electric field $\mathbf{E}$ at point P .



Q3: [40] A spherical charge distribution exists in free space in the region $0 < r' < a$ given by

$$\rho(r') = \rho_0 \left(1 - \frac{r'}{a} \right)$$

- (a) Find the total charge.
(b) Determine $\mathbf{E}$ everywhere.
(c) Determine potential V everywhere.
(d) Sketch both $|\mathbf{E}|$ and V as a function of r .